

What bilinguals tell us about cognitive control: Overview to the special issue

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The current special issue presents the state of the art on the topics of both bilingual language control and executive function, with a particular focus on how bilingualism and cognitive control interact. The contributions to this issue investigate the mechanisms that allow bilinguals to regulate their languages and address how different aspects of language processing might be causally related to cognitive control. Taken together, these papers suggest a more complex engagement and coordination of executive control networks than revealed in past research and a need to more fully characterise those aspects of bilingual language experience that contribute to regulatory processes.

Keywords: Bilingualism; Executive control; Bilingual advantage; Language selection.

Evidence drawn from both behavioural and neurocognitive perspectives supports the idea that bilinguals activate both languages in parallel. A focus of recent research has been to disentangle how the bilingual speaker manages what appears to be an overwhelming feat. If both languages are active, what mechanisms of control are engaged to allow an apparently seamless flow of comprehension and production? At the same time, there appear to be enduring cognitive consequences of bilingualism across the lifespan which suggest that a bilingual’s daily mental juggling between languages may create expertise that enhances domain-general cognitive control.

A critical focus in recent research is to identify the mechanisms that enable bilinguals to select the language they intend to use and to then

understand how these mechanisms might produce the observed consequences of bilingualism for executive control. Many of the past studies have reported correlational findings that make it appealing to suggest that the resolution of cross-language competition might map onto domain-general expertise that produces enhanced cognitive control. Only recently have behavioural and neurocognitive investigations begun to examine how different aspects of language processing might be causally related to cognitive control.

The aim of the current special issue is to present the state of the art on issues of both bilingual language control and executive function with a particular focus on the interplay of bilingualism and cognitive control. To this end, the keynote contribution “Understanding the

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consequences of bilingualism for language processing and cognition” by Kroll and Bialystok provides a critical overview of the research to date on the consequences of bilingualism for cognition and language processing. They argue that bilingual experience has profound consequences for shaping the networks that support both cognition and language. Although much of the past literature has characterised bilingualism categorically (see Luk & Bialystok, this issue) and taken an analytical approach that has been componential, Kroll and Bialystok propose that a multivariate approach is required to reveal the full range of consequences that the use of two languages holds for both cognition and language, to understand the neural processes that support them, and to capture the changes that occur in that network over the lifespan. At a broad level, they claim that the narrative that has been told about bilingualism in the past may only be partly correct, i.e., that the parallel activation of the bilingual’s two languages creates competition that requires resolution and it is that process over the lifespan that is responsible for the observed consequences. At a detailed level, that account may fail to distinguish between different types of bilingual experience that are likely to have specific consequences for how the network that regulates performance is controlled.

The paper “Language control in bilinguals: The adaptive control hypothesis” by Green and Abutalebi, lays out a framework for examining the regulatory network to address just the sort of specificity that Kroll and Bialystok suggest is needed. The adaptive control hypothesis links the way that bilinguals use the two languages to the underlying cognitive and neural control mechanisms that support them. The control mechanisms themselves are hypothesised to change in response to bilingual experience. The idea is to examine the demands that result when particular contexts of bilingual language (e.g., code switching) impose unique demands on control mechanisms. Although the available evidence does not permit a full evaluation of all combinations of contexts and control, the framework provides a promising approach that may ultimately account for the increasingly complex pattern of findings on bilingualism and control processes.

Other papers in the special issue report findings from studies that begin to provide a more nuanced interpretation of the interplay between cognitive processing and control. In “Dual

mechanisms of cognitive control in bilinguals and monolinguals” by Morales, Gómez-Ariza, and Bajo, the performance of bilinguals and monolinguals was compared in the AX-CPT task and in a stop-signal task. The AX-CPT task has been used in the literature on executive function to dissociate proactive and reactive control by differentiating conditions in which it is possible to anticipate a highly probable event from those in which processes that are already engaged must be inhibited. Morales et al. report a bilingual advantage when greater control is required and a differential role for inhibitory processes in bilinguals relative to monolinguals. This work shows that it is necessary to focus on the coordination of more than one component of control to fully account for the observed pattern.

In “Parallel language activation and cognitive control during spoken word recognition in bilinguals”, Blumenfeld and Marian examine the relationship between a language task, spoken word recognition, and a cognitive task, a non-linguistic Stroop task. They report that bilinguals who are more proficient in the L2 reveal greater activation of both languages in the word-recognition task and smaller Stroop effects relative to bilinguals who are less proficient in the L2. A time course analysis also shows that the more proficient bilinguals are able to more quickly resolve cross-language inhibition. Taken together, the results suggest that it is the same control network that is engaged by the language task and the cognitive task.

Bobb and Wodniecka, in “Language switching in picture naming: What asymmetric switch costs (do not) tell us about inhibition in bilingual speech planning”, review the growingly complex literature on cued language switching, a paradigm that has been widely used in the bilingual literature to argue for and against language control through inhibitory mechanisms. Based on methodological considerations, they argue that switch costs are difficult to interpret. In this respect, switch costs by themselves may not be the best index of inhibition in bilingual language control. Instead, they point to alternative indices of inhibition within the paradigm, as well as corroborating evidence from adapted switching paradigms as better ways of assessing the contribution of inhibition to bilingual language control. They call for more systematic research within the switching paradigm that compares task parameters across different bilingual populations.

They also propose that the bilingual advantage in cognitive control may contribute to the observed language switching patterns by interacting with language switching performance.

In “Tracing the bilingual advantage in cognitive control: The role of flexibility in temporal preparation and category switching”, Marzecová, Bukowski, Correa, Boros, Lupiáñez, & Wodniecka provide support for one potential alternative hypothesis of switch costs, which is that bilinguals display increased cognitive flexibility, allowing them to reduce carry-over effects from previous trials in an adapted switching paradigm. In the study, bilinguals switched between two task demands and also between two different timing conditions. For task and timing changes, bilinguals outperformed monolinguals, suggesting that bilinguals may be more effective than monolinguals in resisting the persistent activation of a previous trial.

Luk and Bialystok in “Bilingualism is not a categorical variable: Interaction between language proficiency and usage” argue that assessing the consequences of bilingualism requires a different conception of bilingualism than the categorical notion that is typically standard in research studies today. They specifically look at two variables, language proficiency and language usage, as defined by their Language and Social Background Questionnaire (LSBQ) and how these variables predict the consequences of bilingualism on a verbal and nonverbal task. Consistent with their hypotheses, Luk and Bialystok find that multiple factors best represented bilingual experience in the model while also finding significant correlations between proficiency, usage, and self-rated proficiency.

Prior and Gollan in “The elusive link between language control and executive control: A case of limited transfer” present a novel approach to address the issue of non-linguistic vs. language-specific control mechanisms. They specifically looked at short-term transfer effects of training on language switching and task switching. They compared transfer effects in two groups of bilinguals and a group of monolinguals and found evidence for a limited but significant role of executive control in bilingual language control. They conclude that executive control may play a role in facilitating non-dominant language production and in monitoring to avoid errors.

In “Language conflict in translation: An event related potential study of translation produc-

tion”, Christoffels, Ganushchak, and Koester address the issue of language control in word translation and focus on the temporal dynamics of language control as indexed by ERPs. Their results provide some of the first evidence of how ERPs can be used to understand mechanisms of translation. They argue that certain aspects of language information as available in the input may reduce language competition during translation production.

Together, the studies of this special issue of the *Journal of Cognitive Psychology* illustrate that with respect to language control, bilinguals are not just a one-trick pony – multiple components underlie executive control networks, and each of the many configurations of the bilingual experience uniquely influences the tuning of these control mechanisms. Under certain conditions, regulating language activation may indeed lead to an observed cognitive benefit for bilinguals over monolinguals. But telling a more nuanced story of the consequences of bilingualism on language and cognitive control will require future research in order to disentangle individual components of control while accounting for the complexity of a bilingual’s language experience.

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